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Roll no: 08

Class: BE Computer

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# Assignment No: 01- Boston House Price Prediction
# Linear Regression + Deep Neural Network
# ================================

# Import libraries
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

# Correct Keras import (IMPORTANT FIX)
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Upload dataset
from google.colab import files
uploaded = files.upload()

# Load dataset
boston = pd.read_csv("boston_house_prices.csv")

# Features and target
X = boston[['LSTAT', 'RM', 'PTRATIO']]
y = boston['PRICE']
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=4)
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# Standardization
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# ================================
# Linear Regression
# ================================
lr_model = LinearRegression()
lr_model.fit(X_train_scaled, y_train)
y_pred_lr = lr_model.predict(X_test_scaled)
print("Linear Regression Model Evaluation:")
print("MSE:", mean_squared_error(y_test, y_pred_lr))
print("MAE:", mean_absolute_error(y_test, y_pred_lr))
print("R2 Score:", r2_score(y_test, y_pred_lr))

# ================================
# Neural Network Model
# ================================
model = Sequential([
    Dense(128, activation='relu', input_dim=3),
    Dense(64, activation='relu'),
    Dense(32, activation='relu'),
    Dense(16, activation='relu'),
    Dense(1)
])
model.compile(optimizer='adam', loss='mse', metrics=['mae'])

# Train model
model.fit(X_train_scaled, y_train, epochs=100, validation_split=0.05, verbose=1)

# Evaluate model
y_pred_nn = model.predict(X_test_scaled)
mse_nn, mae_nn = model.evaluate(X_test_scaled, y_test)
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print("\nNeural Network Model Evaluation:")
print("MSE:", mse_nn)
print("MAE:", mae_nn)

# ================================
# Prediction for new data
# ================================
new_data = np.array([[0.1, 10.0, 5.0]])
new_data_scaled = scaler.transform(new_data)
prediction = model.predict(new_data_scaled)
print("\nPredicted House Price:", prediction[0][0])
```

OUTPUT:

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[6] 30s
print("\nPredicted House Price:", prediction[0][0])

Choose files boston_house_prices.csv
boston_house_prices.csv(text/csv) - 35203 bytes, last modified: 11/04/2026 - 100% done
Saving boston_house_prices.csv to boston_house_prices.csv
Linear Regression Model Evaluation:
MSE: 30.340105190234596
MAE: 3.5844321029226935
R2 Score: 0.6733732528519258
Epoch 1/100
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
12/12 ━━━━━━━━━━━━━━━━━━━━ 1s 18ms/step - loss: 564.1665 - mae: 21.8608 - val_loss: 434.4729 - val_mae: 19.7896
Epoch 2/100
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 514.8140 - mae: 20.6610 - val_loss: 376.7899 - val_mae: 18.1815
Epoch 3/100
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 413.3408 - mae: 18.0808 - val_loss: 264.8814 - val_mae: 15.2173
Epoch 4/100
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 249.5218 - mae: 13.7253 - val_loss: 119.3258 - val_mae: 9.8927
Epoch 5/100
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 97.4665 - mae: 7.8985 - val_loss: 54.6319 - val_mae: 5.7517
Epoch 6/100
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 64.3528 - mae: 5.8590 - val_loss: 40.2682 - val_mae: 5.0115
Epoch 7/100
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 44.7926 - mae: 4.9599 - val_loss: 30.9904 - val_mae: 4.6085
Epoch 8/100
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 37.5906 - mae: 4.5127 - val_loss: 26.2615 - val_mae: 3.7423
Epoch 9/100
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 33.8790 - mae: 4.1979 - val_loss: 24.8413 - val_mae: 3.6002
Epoch 10/100
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 31.5361 - mae: 4.0366 - val_loss: 21.3575 - val_mae: 3.4543
Epoch 11/100
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 30.0487 - mae: 3.9407 - val_loss: 19.9678 - val_mae: 3.3629
Epoch 12/100
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 28.9950 - mae: 3.8430 - val_loss: 20.1504 - val_mae: 3.4094
Epoch 13/100
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 27.7556 - mae: 3.7686 - val_loss: 17.9908 - val_mae: 3.2360
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Epoch 100/100
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 13.6384 - mae: 2.4995 - val_loss: 8.9702 - val_mae: 2.5705
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 21.4367 - mae: 2.8299

Neural Network Model Evaluation:
MSE: 21.436683654785156
MAE: 2.829916000366211
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step

Predicted House Price: 83.31667
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